A small clue towards the solution comes from a calculation completed by Stephen Hawking, the physicist famous for having continued to produce outstanding physics despite a medical condition which keeps him confined to a wheelchair and prevents him from speaking without a mechanical aid.

Using quantum mechanics Hawking successfully demonstrated that black holes are always ‘hot’. They emit heat like a stove. It’s the first concrete indication on the nature of ‘hot space’. No one has ever observed this heat because it is faint in the actual black holes which have been observed so far – but Hawking’s calculation is convincing, it has been repeated in different ways, and the reality of the heat of black holes is generally accepted.

The heat of black holes is a quantum effect upon an object, the black hole, which is gravitational in nature. It is the individual quanta of space, the elementary grains of space, the vibrating ‘molecules’ that heat the surface of black holes and generate black hole heat. This phenomenon involves all three sides of the problem: quantum mechanics, general relativity and thermal science. The heat of black holes is like the Rosetta Stone of physics, written in a combination of three languages – Quantum, Gravitational and Thermodynamic – still awaiting decipherment in order to reveal the true nature of time.